

Emergency Room Triage and Patient Flow Database Requirements

Problem Domain:

Within the past five years, Chester County residents have seen the closing of two major ER departments, Brandywine Hospital in 2021 and Jennersville Hospital in 2022. This leaves Chester County, with its 75 cities and just over half a million residents (546.64k), with only two hospitals that have operational ER departments, Chester County and Phoenixville. With only two emergency room departments for the entire Chesco area, Chester County and Phoenixville Hospitals have experienced a huge surge of patients in their ER waiting rooms. The impact of this is that Chester County residents now have longer wait times for triage and treatment, increased patient transfers due to capacity issues and the overworking of doctors and nurses due to higher patient loads, which ultimately affects quality of care.

Due to the increased patient volume in Chester County emergency rooms, I have chosen to design my database, Emergency Room Triage and Patient Flow; to help improve the efficiency and effectiveness of the ER intake process at the Chester County ER department and to find out what areas need improvement. My goal with this database is to streamline the triage process, optimize resource allocation and track patient flow in order to reduce wait times and improve emergency care.

Entity Set Requirements:

Patient:

Each patient has a unique {patient_ID}, name which consists of first and last, date of birth, gender (not required), contact phone number, contact email.

A patient arrives at the ER and is entered into the system.

One patient can have multiple ER visits, but each visit is recorded separately.

Insurance Status:

Each patient can have an insurance status which will consist of a unique {insurance_id} and an insurance status (has insurance, does not have insurance).

Not every patient will have insurance.

Triage Assessment Levels:

Each triage assessment has a unique {triage_id}, time of triage assessment (military time), assigned triage level (ESI_1, ESI_2, ESI_3, ESI_4, ESI_5), a priority status, and description why they came to the ER.

Database will be referencing the Emergency Severity Index (SLI) for triage levels with ESI_1 as immediate and life threatening, ESI_2 as emergent/high risk, ESI_3 as urgent/requires multiple resources, ESI_4 as less urgent/requires one resource and ESI_5 as non-urgent/minimal or no resources needed.

Priority status will be numeric with 5 as highest priority and 1 as the lowest.

Nurses will be conducting all of the triage assessments.

An ESI_1 assessment will not receive a bed assignment, they will go to the fast-track zone and be seen by a doctor there.

Triage levels may not always have patients assigned to them and some triage levels can have multiple patients assigned to it.

Beds:

The ER department has a limited number of beds, each will have a unique {bed_num} and the status of the bed (available, occupied, under maintenance).

A bed can only be occupied by one patient at a time.

Every bed will be assigned to a zone within the ER department.

No bed numbers are duplicated, they are all unique.

Patients who require observation or treatment will be assigned a bed. Not every patient will require observation.

Once a patient is discharged and the room is cleaned, it becomes available again.

Medical Staff:

ER medical staff treat patients and document procedures.

Each medical staff will have a unique {medical_staff_id}, name which consists of first and last, ONE contact number, start of shift and end of shift.

The medical staff consists of doctors and nurses.

Each doctor will be assigned their specialty in the ER (emergency medicine, pediatric emergency medicine, toxicology, ultrasound emergency medicine and disaster medicine).

Not every medical staff will treat a patient due to the fact that a patient may leave before being seen.

Zones:

The ER is divided into zones (Trauma Bay, fast track for minor cases, fast track for critical cases, acute care, intake, triage, pediatric ER, critical care, observation and behavioral health) and patients can be assigned a zone depending on their triage level which is given during triage assessment. Certain beds are assigned to certain zones in the ER department and nurses are then assigned beds.

Zones will consist of zone number (zone one, zone two, zone three, etc) and zone name. So trauma bay will be zone one, fast track for minor cases will be zone two, fast track for critical cases is zone three and so on.

The fast-track zone does not have any beds, only chairs, and it is for patients who have minor issues going on like flu-like symptoms, earaches, etc. but are still seen by a doctor.

Not every patient will be placed into a zone after triage assessment because they may leave before receiving a triage assessment.

Certain beds will be assigned to certain zones within the ER department.

Treatments:

Each ER visit may involve one or more medical procedures (e.g., X-rays, blood tests, IV fluids).

Treatments record procedure type, timestamp, patient outcome and will have a unique {treatment_id} for each patient.

For each treatment with medical staff there will be a role which will describe what medical staff did for a specific treatment.

Transfers:

Due to ER overcrowding, some patients are transferred to other facilities.

Transfers will have unique {transfer_id}, time of transfer, reason for being transferred and receiving hospital.

A patient can only be transferred one time to another hospital.

Er Wait Time:

Calculates several average wait times for patients in the ER.

There will be a unique {wait_time_id}, check in time, triage completion time, bed assignment time, treatment start time, treatment end time, discharge time, left without getting treatment and total wait time.

Not every patient will receive treatment or triage assessment. Some patients may check in and then leave before they can be seen for whatever reason.

ER wait time corresponds with a single ER visit and ER wait time will track several timestamps (check in to triage, triage to bed, etc.). If a patient checks in, they will always have an ER wait time even if they leave before receiving triage.

Relationship Sets:

1. One nurse can conduct many triage assessments per shift. (1:M)
2. There can be one triage assessment conducted per ER wait time. (1:1)
3. One patient will have one ER wait time. (1:M)
4. One patient can be assigned at most one bed per ER wait time. (1:1)
5. Many treatments are performed by many medical staff members. (1:1)
6. One patient can receive multiple treatments. (1:M)
7. Each ER wait time could result in an ER transfer due to hospital overcrowding. (1:1)
8. Each patient can have an insurance status. (1:1)
9. All zones contain multiple beds and are staffed by multiple nurses, with each nurse being assigned responsibility for multiple beds within their designated zone. (M:M)